

Vibhaalakshmi Sivaraman

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<https://vibhaalakshmi.com>

Education	• Massachusetts Institute of Technology	Sep 2019 - Dec 2023
	Ph.D. in Computer Science Advisor: Mohammad Alizadeh	
	• Massachusetts Institute of Technology	Sep 2019
	S.M. in Computer Science Advisor: Mohammad Alizadeh	
	• Princeton University	Jun 2017
	B.S.E. in Computer Science, <i>Summa Cum Laude</i>	
Awards and Honors	• EECS Rising Stars	2023
	• MIT Pillar AI Collective Grant	2023
	• MIT Graduate Women of Excellence Honoree	2023
	• Meta Fellowship Finalist	2021, 2020
	• MIT Jacobs Presidential Fellowship	2017 - 2018
	• Princeton University Computer Science Department Service Award	2017
Industry Experience	• Adept AI Labs and Amazon AGI SF Lab	Jan 2024 - Present
	– Part of research team responsible for developing foundational models for browser-agents that are performant on WebUI understanding and navigation by working across the pre-training and post-training pipeline. – Delivered the first use of RL for Nova ACT’s browser-agent production pipeline: integrated our multimodal model with VeRL, and developed toy datasets relevant to web localization to establish signs-of-life. Experimented with curriculum and multi-task setups with verifiable rewards on browser gym environments, and demonstrated upto 88% improvement in task accuracy with RL training. – Designed datasets to understand when enabling loss on pretraining image tokens benefits downstream image understanding and showed that pretraining with image loss on 25% of the data achieves comparable performance to training without image loss on the full corpus for web localization tasks relevant to Nova ACT. Detailed our findings in a report, sought partners in the broader AGI organization and influenced the addition of image loss in the Nova foundational model’s training recipe. – Worked with product and Alexa Skills team to scope out workflows for new web integrations that included making bookings, changing reservations, and searching for providers and deals across common consumer service platforms.	
	• Google Labs (Teleportation Team)	Oct 2022 - Nov 2023
	Developed alternative neural networks for novel pose reconstruction without optical flow by using attention for super-resolution; model leverages a succinct set of facial feature banks condensed from multiple reference frames in different poses without the overheads of computing attention across all frames; initial evaluation in TensorFlow shows upto 2 dB improvement in PSNR.	

- **Microsoft (Apps and Services Team)** **Summer 2015**
Added features and functionality for tables in Excel's Windows app with more touch-friendly User Interfaces than the Desktop version; implemented UI to append total rows to Excel tables, extended the UI to change table styles; extensively tested features and pushed it to production code
- **AT&T Labs** **Summer 2014**
Designed and implemented a standalone fraud-blocking module with a REST interface that looks up call parameters in a fraud database before placing calls; designed to be independently compatible with all AT&T application servers that process calls

Research Summary

- **Video Conferencing**
 - An end-to-end neural compression system for low-bandwidth video conferencing that uses GAN-based face reconstruction from low-resolution video frames, and is optimized for efficient computation on user devices.
Code: <https://github.com/geminovc>
 - New super-resolution technique for novel pose reconstruction using attention that leverages multiple reference frames.
 - Token-based transformer model to recover from packet loss during video conferences that nearly eliminates all freezes during lossy network conditions.
 - A measurement tool to capture video-conferencing experience in the wild using measurement from a single client's perspective.
 - New rate control mechanism for real-time video that responds to bandwidth fluctuations faster by adding padding traffic during video 'off' periods, and controlling the encoder target bitrate on short timescales.
- **Video Streaming**
 - *SRVC*, a new codec design that augments existing codecs with a small, content-adaptive super-resolution model that significantly boosts video quality.
Code: <https://github.com/AdaptiveVC/SRVC>
 - *Minerva*, an end-to-end transport protocol optimized for QOE fairness across users streaming video over a shared bottleneck.
- **Credit Network Throughput**
 - *Spider*, a new network architecture and routing protocol, inspired by traditional data networks, that achieves high-throughput routing in credit networks.
Code: <https://github.com/spider-pcn>
 - Theoretical understanding of the effect of topology on credit network throughput and practical guidance for topology design.
- **Heavy Hitter Detection**
Hashpipe, an algorithm that uses a pipeline of hash tables to detect large flows in a network at line-rate on commodity switch hardware.
Code: <https://github.com/vibhaa/hashpipe>

Pre-prints

- **Multi-Resolution Multi-Reference Talking Head Synthesis via Implicit Warping**
Vibhaalakshmi Sivaraman, Xuan Luo, Anne Menini, Mohammad Alizadeh, Rahul Garg
https://people.csail.mit.edu/vibhaa/files/gemino_attn.pdf

- **Vidaptive: Efficient and Responsive Rate Control for Real-Time Video on Variable Networks**
Pantea Karimi, Sadjad Fouladi, **Vibhaalakshmi Sivaraman**, Mohammad Alizadeh
<https://arxiv.org/abs/2309.16869>
- **Reparo: Loss-Resilient Generative Codec for Video Conferencing**
Tianhong Li, **Vibhaalakshmi Sivaraman**, Lijie Fan, Mohammad Alizadeh, Dina Katabi
<https://arxiv.org/abs/2305.14135>

Publications

- **Gemino: Practical and Robust Neural Compression for Video Conferencing**
Vibhaalakshmi Sivaraman, Pantea Karimi, Vedantha Venkatapathy, Mehrdad Khani, Sadjad Fouladi, Mohammad Alizadeh, Frédo Durand, Vivienne Sze
USENIX NSDI 2024
- **Efficient Video Compression via Content-Adaptive Super-Resolution**
Mehrdad Khani, **Vibhaalakshmi Sivaraman**, Mohammad Alizadeh.
IEEE ICCV 2021
- **The Effect of Network Topology on Credit Network Throughput**
Vibhaalakshmi Sivaraman, Weizhao Tang, Shaileshh Bojja Venkatakrishnan, Giulia Fanti, Mohammad Alizadeh.
IFIP Performance 2021
- **High Throughput Cryptocurrency Routing in Payment Channel Networks**
Vibhaalakshmi Sivaraman, Shaileshh Bojja Venkatakrishnan, Kathleen Ruan, Parimarjan Negi, Lei Yang, Radhika Mittal, Giulia Fanti, Mohammad Alizadeh.
USENIX NSDI 2020
- **End-to-End Transport for Video QOE Fairness**
Vikram Nathan, **Vibhaalakshmi Sivaraman**, Ravichandra Addanki, Mehrdad Khani, Prateesh Goyal, Mohammad Alizadeh.
ACM SIGCOMM 2019
- **Routing Cryptocurrency with the Spider Network**
Vibhaalakshmi Sivaraman, Shaileshh Bojja Venkatakrishnan, Mohammad Alizadeh, Giulia Fanti, Pramod Viswanath.
ACM HotNets 2018
- **Heavy-Hitter Detection Entirely in the Data Plane**
Vibhaalakshmi Sivaraman, Srinivas Narayana, Ori Rottenstreich, S.Muthukrishnan and Jennifer Rexford.
ACM SOSR 2017

Presentations

- **Gemino: Practical and Robust Neural Compression for Video Conferencing**
 - UC Berkeley NetSys, Berkeley, February 2023
 - Stanford Systems Seminar, Palo Alto, February 2023
 - Microsoft Research, Redmond, November 2022
- **Designing Credit Networks for High Throughput**

- UC Berkeley Systems Seminar, Berkeley, April 2021
- Rutgers University Systems Seminar, New Brunswick, March 2021

- **High-Throughput Cryptocurrency Routing in Payment Channel Networks**

- USENIX NSDI, Santa Clara, February 2020
- The Lightning Conference, Berlin, October 2019
- MIT Digital Currency Initiative, Cambridge, October 2019
- Carnegie Mellon CyLab Distinguished Seminar, Pittsburgh, September 2019
- ACM Workshop on Hot Topics in Networks, Redmond, November 2018

- **Heavy Hitter Detection Entirely in the Data Plane**

- ACM Symposium on SDN Research, Santa Clara, April 2017
- Cornell University, Ithaca, February 2017
- New England Networking and Systems Day, Boston, October 2016

**Teaching
Experience**

- **Teaching Assistant, MIT EECS** **Spring 2019**
6.858: Computer Systems Security
Overall Rating: 6.3/7.0

- **Undergraduate Lab Teaching Assistant, Princeton University** **2015 - 2017**
COS 126: General Computer Science
COS 226: Algorithms and Data Structures
COS 217: Introduction to Programming Systems
Received Student Teaching Award from the Computer Science Department

- **Peer Tutor, Princeton University** **2015 - 2017**
MAT 201: Multivariable Calculus
MAT 202: Linear Algebra
COS 126: General Computer Science
Tutored as a peer tutor under the McGraw Center for Teaching and Learning

**Service
Experience**

- MIT EECS Resource for Easing Friction and Stress **2020 -2023**
- Hall Councilor for Sidney Pacific Graduate Dorm **2018 - 2021**
- MIT Graduate Women in Course 6 (GW6) Co-President **2018**
- Princeton Women in Computer Science (PWICS) Co-President **2016 - 2017**